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Brexit: Impacts on housing markets across Europe

Abstract:

In this paper I aim to analyse how Brexit affected the Housing Markets in the EU and UK and if there one region was impacted more than the other. The motivation behind this research is that Brexit and housing are currently two important topics in Europe so I wanted to look into them more and see if there were any significant trends related to the two. To conduct this research, I used OLS analysis and comparisons. I found through this research that Brexit had little effect on the EU, and a much larger effect on the UK's housing market, with some inverse trends across some indicators.

Introduction:

In my research I intended to look at the EU and UK housing markets before and after the vote to leave the EU was confirmed. This is an important topic as both Brexit and concerns about housing are currently two very heated and modern issues in the UK right now (as well as many other economic-based issues), with impacts across all of Europe. My goal with my research was to look at how impactful the vote truly was between the two Unions, and to identify if they were affected equally or not. Thus, I wanted to know: How did the vote to leave the EU affect the Housing Markets in the EU and UK? I chose the vote instead of the official Brexit date due to the fact that after the vote was confirmed, the UK began implementing policies to make the official break easier and more fluid, thus Brexit itself had

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no real treatment effect. To do this research, I decided upon investigating several general economic indicators such as CPI, interest rates, as well as more housing-specific indicators such as the number of new builds in comparison the year prior, and the housing index, to name a few. Economic variables tend to be good indicators of how well a nation (or union) is doing overall, and these types of variables would directly impact housing prices and affordability. My research question is motivated primarily by the controversial nature of housing policies in the UK that tend to favour immigrants and refugees, as well as the controversiality surrounding the vote to leave the EU, which was won with only a 52% majority. That being said, I hypothesise that the vote caused short term negative effects on the housing market in both the UK and EU but I also hypothesise that both unions will return to their prior housing market states in the long-run.

Literature Review:

After reading many academic articles on housing issues, I noticed that the Housing cycles in the UK and the UK housing bubble case study analysis articles provide background on understanding housing market cycles and bubbles in general, but don't directly address the impact of Brexit. The impact of immigration on UK house prices article is indirectly related but might be a relevant factor to consider when investigating the impact of Brexit on the UK housing market given the idea that Brexit has led to reduced immigration to the UK, which could affect housing demand. Furthermore, the impact of Brexit on the UK housing market article discusses a potential housing boom due to Brexit. However, the viewpoint of the article is from 2021, and the long-term impact of Brexit on the housing market is still unclear, which makes sense given the limited amount of time post the official date of Brexit in 2020. The existing literature provides context for understanding the UK housing market and the factors that can influence it. However, there is a lack of studies that directly investigate the

causal impact of Brexit on the housing markets of the UK and the EU. Moreover, there is very limited research into the effects of the Brexit vote, only the true date of Brexit. As such, through my research, I aim to fill a gap in the existing literature by examining the causal impact of Brexit on the housing markets of both the UK and the EU using the Brexit vote as a treatment variable instead of brexit. This is an important topic to investigate as Brexit is a major economic and political event that could have significant consequences for housing markets, yet there is little studies on how the EU and UK were impacted in relation to one another, and neglects the fact that policies were being implemented before the official Brexit date that were meant to start working towards the goal of the separation from the EU.

Data and Descriptive Statistics:

Within my dataset, I have panel data for 36 total observations across 7 different variables and 2 groups (the UK and the EU). I have so few observations due to the complications with the underreported nature of UK statistics, and due to the time which I am confined to. If I were to continue this research, I would look into adding more observations to make my findings more robust. The data is quarterly data from 2011-2019. I would have preferred monthly data with more years however this was not possible from what I was able to find. Furthermore, all of the variables are observed twice - once for the UK and once for the EU. Since I do not combine the data for each group, GDP is measured in Great British Pounds and Euros, respectively. These observations were selected based purely on how much data was available for any given variable with relation to housing markets for both the EU and UK. Despite this, I do not believe there is potential for much selection bias, due to the fact that the complications with the data collection was that the UK tends to not publicly produce data that is largely incomplete, inaccurate, or has the potential to harm individual privacy. My data was sourced from several places including: Ourworldindata, FRED, the UK

Housing Observatory, ons.gov.uk (the UK government statistics website), the Bank of England, and ecb.europa.eu (the EU central bank).

Looking at the data closely, the variables I use in my research are: Number of New Builds compared to the prior year, Consumer Price Index, Population, Housing Index, Average Hourly Wages of Manufacturing Workers, Interest Rates, and GDP. All of the indexes are indexed to the specific date of the vote to leave the EU: 23/07/2016. The Number of New Builds is a measure of how many permits were issued per year for the purpose of construction of new residential buildings by private entities. The Consumer Price Index is a measure of the average price of goods paid by consumers for all goods. Population is a raw value of the number of people living within the region, with the region being either the EU or UK in this instance. The Housing Index is an overall measure of the affordability, stability, and accessibility of houses. The Average Hourly Wages of Manufacturing Workers is a measure of the national average wage received by anyone working in the manufacturing industry. Interest Rates is the percent at which the central bank had set the interest rate. GDP is a measure of the overall well-being of the region's economic well-being.

date	UKNB	EUNB	UKCPI	EUCPI	UKPop	
Length:39	Min. : -769.20	Min. : -824.34	Min. : 33.65	Min. : 100	Min. : 40472000	
Class :character	1st Qu.: -199.90	1st Qu.: -95.84	1st Qu.: 141.27	1st Qu.: 1092	1st Qu.: 40610000	
Mode :character	Median : -130.53	Median : -35.80	Median : 208.33	Median : 1242	Median : 40986500	
	Mean : -95.30	Mean : 12.75	Mean : 194.03	Mean : 1297	Mean : 41008389	
	3rd Qu.: -27.97	3rd Qu.: 89.86	3rd Qu.: 239.66	3rd Qu.: 1554	3rd Qu.: 41361750	
	Max. : 1495.59	Max. : 793.18	Max. : 387.50	Max. : 2117	Max. : 41631000	
	NA's : 3	NA's : 3	NA's : 3	NA's : 3	NA's : 3	
EUPop	UKHI	EUHI	UKHWI	EUHWI	UKIR	EUIR
Min. : 326636541	Min. : -31.12	Min. : 92.67	Min. : 89.48	Min. : 88.82	Min. : 0.2500	Min. : -0.5000
1st Qu.: 328081093	1st Qu.: 14.33	1st Qu.: 94.11	1st Qu.: 90.99	1st Qu.: 93.44	1st Qu.: 0.5000	1st Qu.: -0.4000
Median : 328977521	Median : 54.82	Median : 96.59	Median : 92.44	Median : 98.00	Median : 0.5000	Median : -0.2000
Mean : 328776117	Mean : 51.64	Mean : 99.65	Mean : 92.30	Mean : 98.24	Mean : 0.5069	Mean : -0.1597
3rd Qu.: 329602771	3rd Qu.: 84.18	3rd Qu.: 104.57	3rd Qu.: 93.47	3rd Qu.: 102.50	3rd Qu.: 0.5000	3rd Qu.: 0.0000
Max. : 330706811	Max. : 132.55	Max. : 115.05	Max. : 95.28	Max. : 109.48	Max. : 0.7500	Max. : 0.7500
NA's : 3	NA's : 3	NA's : 3	NA's : 3	NA's : 3	NA's : 3	NA's : 3
UKGDP	EUGDP	Year	Treat	QToTreat		
Min. : 412364	Min. : 3256446	Min. : 2011	Min. : 0.0000	Min. : -23.00		
1st Qu.: 441142	1st Qu.: 3288082	1st Qu.: 2013	1st Qu.: 0.0000	1st Qu.: -14.25		
Median : 478258	Median : 3416231	Median : 2015	Median : 0.0000	Median : -5.50		
Mean : 483250	Mean : 3445679	Mean : 2015	Mean : 0.3611	Mean : -5.50		
3rd Qu.: 523382	3rd Qu.: 3596268	3rd Qu.: 2017	3rd Qu.: 1.0000	3rd Qu.: 3.25		
Max. : 565758	Max. : 3720732	Max. : 2019	Max. : 1.0000	Max. : 12.00		
NA's : 3	NA's : 3	NA's : 3	NA's : 3	NA's : 3		

Looking at the summary statistics, it is worth making a few notes on some variables.

Firstly, the “date” variable is used to determine which quarter of which year the row of data is

analysing. The “QtoTreat” variable is a simple counter with Quarter 4 of 2016 being equal to zero. It is a variable I intended to use if I had more time to look at the UK and the Brexit vote as an event study. Finally, it should be noted that the “Treat” variable is a dummy variable indicating whether or not the vote to leave the EU had occurred, where post-vote = 1. As such, we are able to ignore these variables when reading the summary statistics, as well as the “Year” variable.

Taking a look at some of the most notable figures to look at from this chart, the population values should be taken into consideration. It is clear that the UK has fewer people, but only marginally so when also accounting for geography. Thus, the houses are most likely more densely spread out in the UK, which may impact my results. It is also worth noting that the EU has more variation in interest rates, ranging from -0.5% to 0.75%, as compared to the UK’s 0.25% to 0.75%. This could mean the ECB is more responsive to short-term shocks as compared to the Bank of England, also leading to potential room for error in our results. It should lastly be noted that the UK has on average a higher GDP than the EU, suggesting that overall the UK is more likely to withstand a short-term shock with less of an impact to the overall economy as compared to the EU.

Econometric Model and Predictions:

When creating my model for analysis, I used OLS regressions to look at the effects of specific indicators on the housing markets in the UK and EU. The model I used was: $\text{Housing Index} = \alpha + \beta_1(\text{New Builds}) + \beta_2(\text{CPI}) + \beta_3(\text{Interest Rates}) + \beta_4(\text{Brexit Vote})$. I used two models, both identical. One was for analysing the UK, one was for analysing the EU. The response variable is the housing index - a measure of the overall affordability, price, condition, etc of the average house in the given region. I use the number of permits issued for private institutions to construct new residential buildings as my primary explanatory variable

of interest. From there, I also use the consumer price index (a measure of the average price of goods paid by consumers for all goods in a given region), interest rates set by the central bank of each region, and a dummy variable to determine if the vote for the UK to leave the EU had occurred yet. This is an appropriate strategy of analysis because OLS models are useful for making predictions and for looking at potential causal relationships. If I were to continue this research, I would develop the model further and then use it to make predictions on the future of the housing markets in the UK and EU to see if they are predicted to trend similarly or not moving forward.

My primary explanatory variable of interest is the number of new builds. I used this as my primary variable because if there are many private contractors requesting and receiving permits for new construction, it most likely indicates that the housing market is either booming or undergoing serious changes, making it a significant variable to study. It allows me to make a causal argument because not only is it a good determiner of where the market supply is, but also gives a foresight to where prices are most likely going to be in the future. For example, if the market is ripe now, then more permits will be issued, leading to more total houses and a downward pressure on housing prices, making them more affordable. This is true assuming that all of the permits issued are followed through with successful completion of building the residential building, and that the suppliers are building these new buildings to meet demand.

This is potentially a weak approach due to the fact that it determines a linear trend, whereas the true housing market may be more complex. Furthermore, it is possible that many key indicators are not included in my model, thus skewing my results further. There are many threats to my identification strategy. These include omitted variable bias, in that there may be some variable not included in my limited model, as well as measurement bias since the data was collected from so many different sources. There may also be concerns about

non-linearity in which the true relationship I am studying is not linear in nature, something that is most likely true in the case of the housing markets. The small sample size would raise concerns regarding the validity of my results, as well as the potential lack of upholding the parallel trends assumption (stemming from a lack of a control group).

An ideal strategy for analysing how the Brexit vote differently impacted the EU and UK housing market's would be to include more variables, such as specific policy changes. I would also use a lot more observations, with monthly data for at least double the number of years. I would then look at a difference-in-difference analysis to properly quantify the effect of the Brexit vote. Furthermore, I would also do an Event Study looking at the EU and the UK separately to visually look at the true effect of the Brexit vote independent of region (region being the UK and EU). Finally, an ideal model would compare two regions of identical traits, holding one constant as a control. I would need 4 total regions, with two identical UKs and two identical EUs such that I may compare the effect of the vote on the EU and UK properly. My model falls short due to a lack of data and therefore an inability to produce a difference-in-differences analysis nor a proper event study. My model also does not account for specific policy changes, and is comparing two regions that are similar but not identical. Moreover, I do not have a control variable to compare the treatment effect to.

I would predict that my model will potentially overestimate the coefficient estimates, however I believe it will still be somewhat accurate in terms of proportional effect. Meaning I believe while the true magnitude of the coefficients may differ from my model's prediction, the coefficients themselves will be represented well in terms of their strength of effect in comparison to one another. More specifically, I believe that as more new building permits are issued, the housing index should decrease meaning lower affordability despite greater access, as a flooding of the housing market with new builds will increase the average price of a house, stopping less privileged families from purchasing these new developments. I believe

the same would be true for interest rates due to reduced consumer spending, meaning less customers for homes.

Results:

UK Model Results		EU Model Results	
Dependent variable:		Dependent variable:	
UKHI		EUHI	
UKCPI	-0.393*** (0.048)	EUCPI	-0.0003 (0.002)
UKNB	-0.016 (0.014)	EUNB	-0.001 (0.002)
UKIR	-116.004*** (32.893)	EUIR	-1.403 (3.209)
Treat	8.472 (9.835)	Treat	11.929*** (1.747)
Constant	182.019*** (19.660)	Constant	95.521*** (2.200)
Observations	36	Observations	36
R2	0.715	R2	0.790
Adjusted R2	0.679	Adjusted R2	0.763
Residual Std. Error	26.560 (df = 31)	Residual Std. Error	3.377 (df = 31)
F Statistic	19.469*** (df = 4; 31)	F Statistic	29.114*** (df = 4; 31)
Note: *p<0.1; **p<0.05; ***p<0.01		Note: *p<0.1; **p<0.05; ***p<0.01	

When looking at the results, it is important to first interpret each important value. In the case of the UK, we see the following significant trends.

For each 1 unit increase on the UK Consumer Price Index, we expect the UK Housing Index to decrease by 0.39 units on average, assuming there are the same number of permits issued as the prior year, and the interest rate is constant. This is statistically significant at the 1% level, and is important economically too. This is because we know that as overall prices increase within the UK, we expect houses overall to become less affordable and less accessible, worsening any effect an increase in prices may have.

For each 1 more permit issued as compared to the prior year, we expect the UK Housing Index to decrease by 0.02 units on average assuming the Consumer Price Index and Interest Rates are held constant. This is not statistically significant at any standard level, and it also does not give much insight economically either, despite my predictions.

For each 1% increase in Interest Rates in the UK, we expect the UK Housing Index to decrease by 1.16 units on average assuming the number of permits issued and Consumer Price Index are held constant. This is statistically significant at the 1% level, and also has a large economic impact. This can be used to help reform the UK housing market through policy change in the future. In the case of the EU, we see similar trends, with similar economic impacts.

For each 1 unit increase on the EU Consumer Price Index, we expect the EU Housing Index to increase by 0.02 units on average assuming there are the same number of permits issued as the prior year, and the interest rate is constant.

For each 1 more permit issued as compared to the prior year, we expect the EU Housing Index to decrease by 5.59 units on average assuming the Consumer Price Index and Interest Rates are held constant.

For each 1% increase in Interest Rates in the EU, we expect the EU Housing Index to decrease by 6.22 units on average assuming the number of permits issued and Consumer Price Index are held constant. While none of the EU model's coefficients are statistically significant, this could be due to the small sample size being analysed, thus not creating robust statistical outcomes. My results are consistent with findings from other papers in general, however the magnitude is not, as well as the lack of statistical significance for most of my explanatory variables. I believe this is due to the small sample size used.

Conclusion:

After analysing the effect of the brexit vote on both the UK and EU housing markets, using OLS regression models utilising variables such as number of new building permits, consumer price index, and interest rates, I found that overall, the UK housing market seemed to be highly impacted by the brexit vote given the higher magnitude of the estimated

coefficients. I also found that despite seeing negative relationships between all three variables and the housing index in both countries, the brexit vote itself was estimated to have a positive relationship. This could be an issue known as Simpson's Paradox, also referred to as omitted variable bias, in which we are observing reversed trends due to the lack of including a certain key variable in the model. Nonetheless, these results are important since they can be used to help implement reforms in policy in both regions, but we know that the UK reforms would need to be stronger and more impactful overall. The results are also important as we see that interest rates were the most important factor on the housing markets for both regions, so when issuing reforms we know that the EU and UK should focus on making changes to the interest rates, assuming the rest of the economy is positively impacted by the chosen reforms. Despite this, we also know that this model had some issues. For instance, we observed omitted variable bias, small sample sizes, and largely statistically insignificant coefficients. Furthermore, there were issues with non-linearity, in the sense that I applied a linear model to a potentially non-linear relationship. In the future, it would be interesting to look at how population changes would impact the housing markets?, and to look at whether the supply met the demand efficiently throughout time in the housing markets?

References:

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Appendix:

```
```{r setup, include=FALSE}

knitr::opts_chunk$set(echo = TRUE, message=FALSE, warning=FALSE, fig.height = 3,
fig.width = 6)

library(tidyverse)

library(ggplot2)

library(stargazer)

library(corrplot)

```

#Load Data

```{r}

EUUK <- read_csv("~/Desktop/481CSV3.csv")

head(EUUK)

```

#Summary Statistics

```{r}

summary(EUUK)

```

```{r}

ggplot(EUUK, aes(Year, UKNB, colour = Treat)) +

 geom_point() +

 geom_smooth(se = FALSE) +

 ggtitle("UK New Builds")

ggplot(EUUK, aes(Year, EUNB, colour = Treat)) +

 geom_point() +
```

```

 geom_smooth(se = FALSE) +

 ggtitle("EU New Builds")
 ...

 ``{r}

 ggplot(EUUK, aes(Year, UKHI, colour = Treat)) +

 geom_point() +

 geom_smooth(se = FALSE) +

 ggtitle("UK Housing Index")

 ...

 ``{r}

 ggplot(EUUK, aes(Year, EUHI, colour = Treat)) +

 geom_point() +

 geom_smooth(se = FALSE) +

 ggtitle("EU Housing Index")
 ...

 ``{r}

 ggplot(EUUK, aes(Year, UKPop, colour = Treat)) +

 geom_point() +

 geom_smooth(se = FALSE) +

 ggtitle("UK Population")

 ...

 ``{r}

 ggplot(EUUK, aes(Year, EUPop, colour = Treat)) +

 geom_point() +

 geom_smooth(se = FALSE) +

```

```
ggtitle("EU Population")
```

```
```
```

```
```{r}
```

```
ggplot(EUUK, aes(Year, UKCPI, colour = Treat)) +
```

```
 geom_point() +
```

```
 geom_smooth(se = FALSE) +
```

```
 ggtitle("UK CPI")
```

```
ggplot(EUUK, aes(Year, EUCPI, colour = Treat)) +
```

```
 geom_point() +
```

```
 geom_smooth(se = FALSE) +
```

```
 ggtitle("EU CPI")
```

```
```
```

```
```{r}
```

```
ggplot(EUUK, aes(Year, UKIR, colour = Treat)) +
```

```
 geom_point() +
```

```
 geom_smooth(se=FALSE)+
```

```
 ggtitle("UK Interest Rates")
```

```
ggplot(EUUK, aes(Year, EUIR, colour = Treat)) +
```

```

geom_point() +

geom_smooth(se=FALSE)+

ggtitle("EU Interest Rates")
'''

'''{r}

ggplot(EUUK, aes(Year, UKHWI, colour = Treat)) +

geom_point() +

geom_smooth(se = FALSE) +

ggtitle("UK Hourly Manufacturing Wages")

ggplot(EUUK, aes(Year, EUHWI, colour = Treat)) +

geom_point() +

geom_smooth(se = FALSE) +

ggtitle("EU Hourly Manufacturing Wages")
'''

'''{r}

ggplot(EUUK, aes(Year, UKGDP, colour = Treat)) +

geom_point() +

geom_smooth(se = FALSE) +

```

```
ggtitle("UK GDP")
```

```
ggplot(EUUK, aes(Year, EUGDP, colour = Treat)) +
```

```
geom_point() +
```

```
geom_smooth(se = FALSE) +
```

```
ggtitle("EU GDP")
```

```
```\n
```

```
```\n{r}\n
```

```
UK <- lm(formula = UKHI ~ UKCPI + UKNB + UKIR + Treat, data=EUUK)
```

```
EU <- lm(formula = EUHI ~ EUCPI + EUNB + EUIR + Treat, data=EUUK)
```

```
```\n
```

```
#Model Output
```

```
```\n{r}\n
```

```
stargazer(UK, type = "text", title = "UK Model Results")
```

```
```\n
```

```
```\n{r}\n
```

```
stargazer(EU, type = "text", title = "EU Model Results")
```

```
```\n
```